4 DE AGOSTO 2014

1) PROPIEDAD: $n\,x[n] \longrightarrow -z\,\frac{dX(z)}{dz}$

a) $X(z) = \log(1-2z)\,; \quad |z| < 0{,}5$

$$\frac{dX(z)}{dz} = \frac{(-2)}{1-2z}$$

$$\frac{-dX(z)}{dz} = \frac{2}{1-2z} \Rightarrow \frac{2}{z(z^{-1}-2)} = \frac{2}{-2z(1-\frac{1}{2}z^{-1})}$$

$$\frac{-2\,dX(z)}{dz} = \frac{-1}{1-\frac{1}{2}z^{-1}}$$

$$\Rightarrow n\,x[n] = +\left(\frac{1}{2}\right)^n u[-n-1]$$

$$x[n] = \frac{1}{n}\left(\frac{1}{2}\right)^n u[-n-1]$$

$$x[n] = \frac{2^{-n}}{n}\,u[-n-1]$$

b) $X(z) = \log(1-2z^{-1}) \qquad |z| > 2$

$$\frac{dX(z)}{dz} = \frac{(-2)(-1)z^{-2}}{1-2z^{-1}} = \frac{2z^{-2}}{1-2z^{-1}} = \frac{1}{z}\,\frac{2z^{-1}}{1-2z^{-1}}$$

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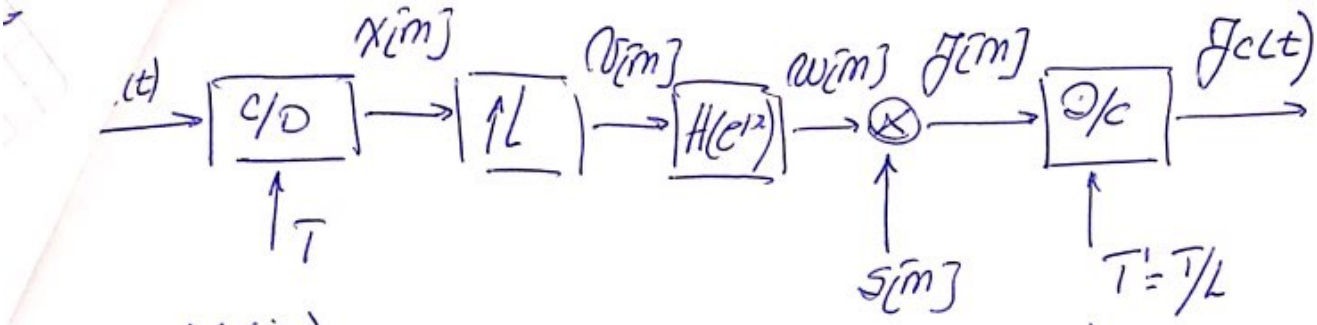
$$z\frac{dX(z)}{dz} = \frac{2z^{-1}}{1-2z^{-1}} \Rightarrow -z\frac{dX(z)}{dz} = \frac{-2z^{-1}}{1-2z^{-1}}$$

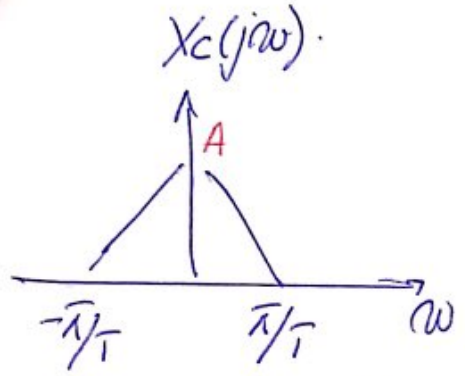
$$\Rightarrow n\,x[n] = (-2)\,2^{n-1}\,u[n-1]$$

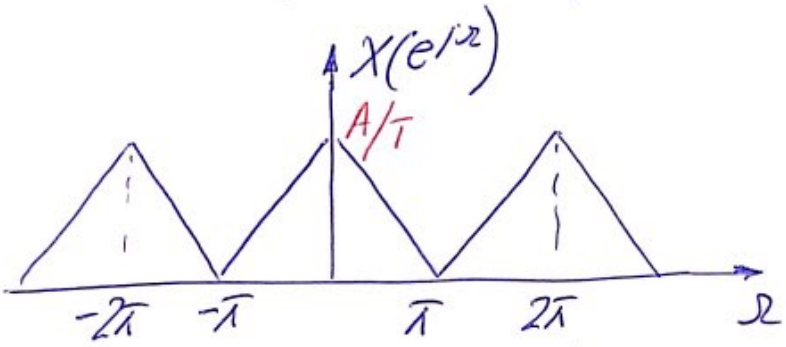
$$x[n] = \frac{1}{n}(-1)\,2^{n}\,u[n-1]$$

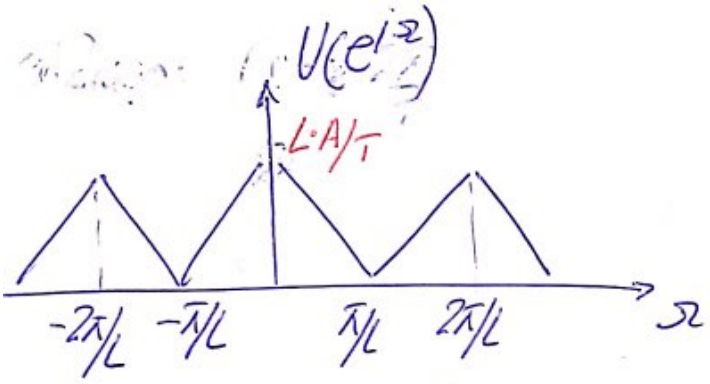
$$x[n] = (-1)\frac{2^n}{n}\,u[n-1]$$

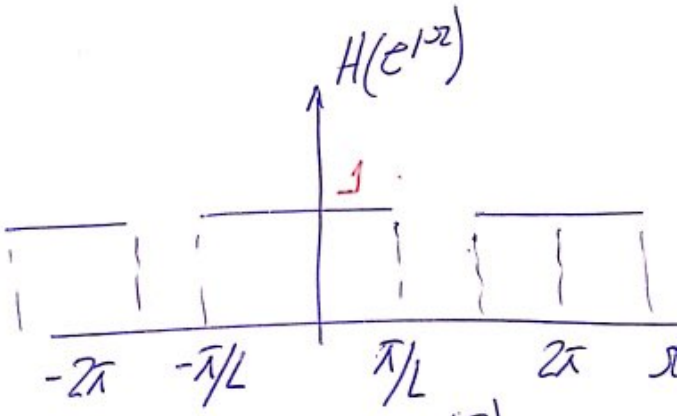
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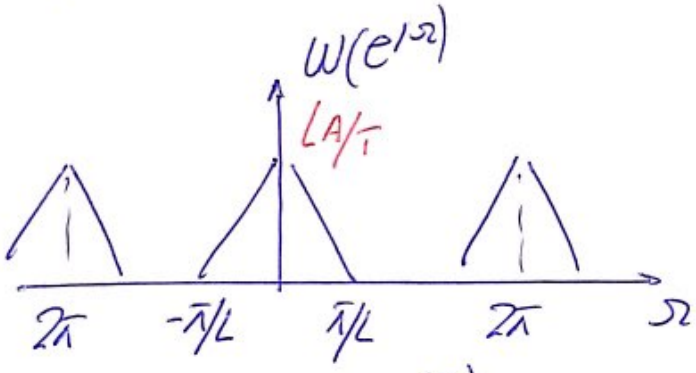




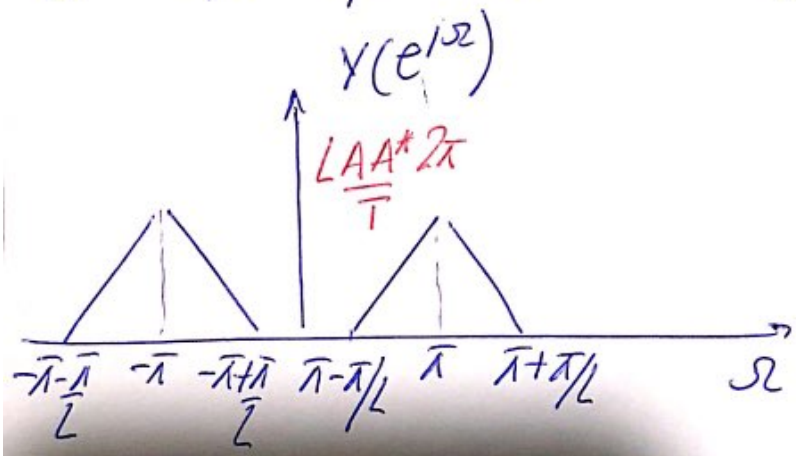






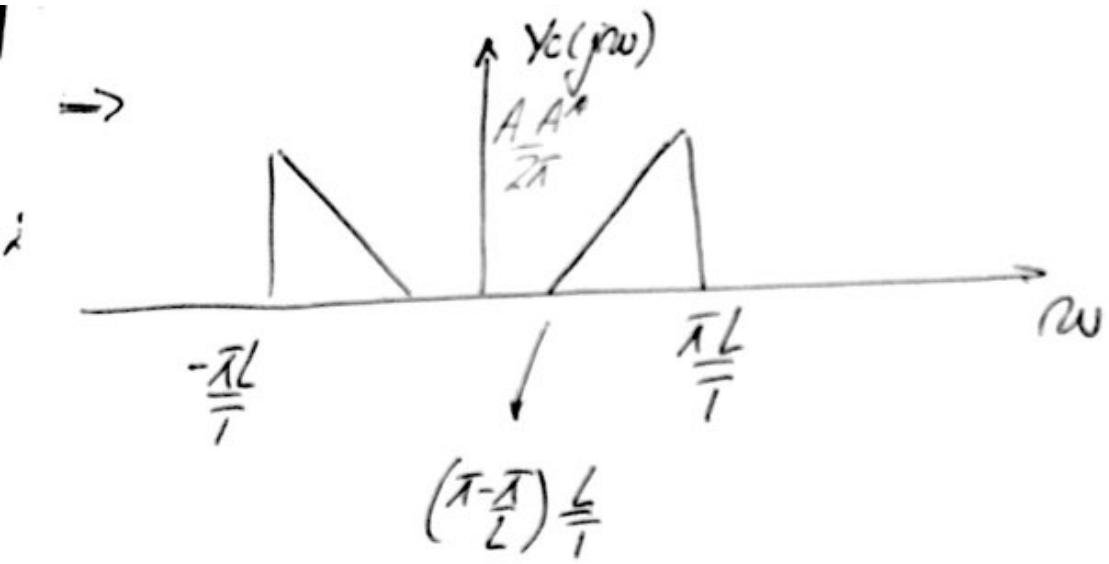






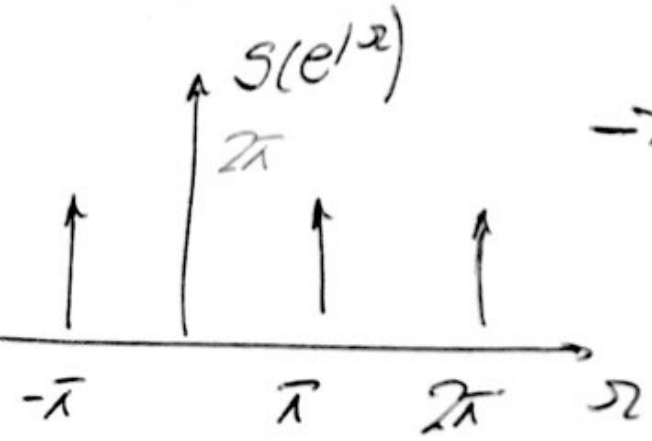
$$y[m] = s[m] \cdot w[m] \Rightarrow Y(e^{j\Omega}) = \frac{1}{2\pi} S(e^{j\Omega}) * W(e^{j\Omega})$$

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$$\Rightarrow \frac{\pi L}{I} = \frac{3\pi}{I} \Rightarrow \boxed{L = 3}$$

$$\Rightarrow \frac{A A^*}{2\pi} = A \Rightarrow \boxed{A^* = 2\pi}$$



$$\rightarrow \boxed{s[n] = e^{j\pi n}}$$

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